

chap_simples_comparaison

2025-07-09

Étude de la stratégie (SI, HT) par simulations de Monte-Carlo

```
library(sampling)
data(rec99)

N <- nrow(rec99)
n <- 70
n.sim <- 10000

est.si <- numeric(n.sim)
piks <- rep(n/N,n)
set.seed(123456)
for (i in 1:n.sim)
{
  si.rec <- srswor(n,N)
  ys <- rec99$LOGVAC[si.rec==1]
  est.si[i] <- HTestimator(ys,piks)
}

mean(est.si)

## [1] 10799.5
# [1] 10757.02
(cv.si <- sd(est.si)/mean(est.si))

## [1] 0.1896896
# [1] 0.1918109

ty <- sum(rec99$LOGVAC)
(mean(est.si)-ty)/ty

## [1] 0.002925289
# [1]-0.001020021
```

Étude de la stratégie (BE, HT) par simulations de Monte-Carlo

```
BE1 <- function(N,Pi)
{
  u <- runif(N, 0, 1)
  as.numeric(u<Pi)
}

est.be <- numeric(n.sim)
n.be <- numeric(n.sim)
set.seed(123456)
```

```

for (i in 1:n.sim)
{
  s.be <- BE1(N,n/N)
  n.be[i] <- sum(s.be==1)
  rec.be <- rec99[s.be==1,]
  est.be[i] <- (N/n)*sum(rec.be$LOGVAC)
}
mean(est.be)

## [1] 10793.04

# [1] 10764.34
(cv.be <- sd(est.be)/mean(est.be))

## [1] 0.2189063

# [1] 0.2193195
(mean(est.be)-ty)/ty

## [1] 0.002325544

# [1]-0.000339896

quantile(n.be,probs=c(0.025,0.1,0.5,0.9,0.95,0.975))

## 2.5% 10% 50% 90% 95% 97.5%
## 55 60 70 80 83 85

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```

Étude de la stratégie (SY, HT) par simulations de Monte-Carlo

```

pik <- rep(n/N,N)
est.sy <- numeric(n.sim)
set.seed(123456)
for (i in 1:n.sim)
{
  sy.rec <- UPsystematic(pik)
  ys <- rec99$LOGVAC[sy.rec==1]
  piks <- pik[sy.rec==1]
  est.sy[i] <- HTestimator(ys,piks)
}
mean(est.sy)

## [1] 10766.32

# [1] 10774.91
(cv.sy <- sd(est.sy)/mean(est.sy))

## [1] 0.34274

# [1] 0.1812545
(mean(est.sy)-ty)/ty

## [1] -0.0001561964

# [1] 0.0006417162

```

```
orderlog <- order(rec99$LOG)
rec.tri <- rec99[orderlog,]
```